

Quiz

Name : _____

October 24

1. Consider some function $F(x, y)$

$$F(x, y) = A$$

Proposition 1 on page 295 says that if $\frac{\partial F(x_0, y_0)}{\partial y} \neq 0$, then the equation has a unique local solution.

Does $F(x, y) = x^{1/2}y^{1/2} = 5$ have a unique local solution at $(1, 1)$?

2. True or False?

- (a) The Jacobian and the Hessian must be a square matrix.
- (b) The Jacobian and the Hessian must be symmetric.
- (c) A matrix is invertible so long as its determinant is not 0.
- (d) The Hessian is used to test if a multivariate function is concave or convex.
- (e) It matters for mixed derivatives whether you first take the derivative of a smooth function with regards to the first or second argument.